

Retail Customer Retention Analytics – TARGET

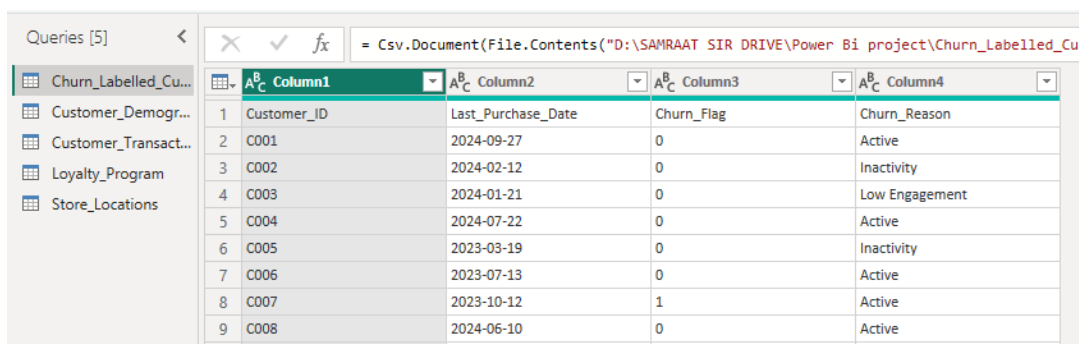
Approach Sheet

Task 1: Data Modelling & Cleaning

- Load and transform datasets in Power Query

To load the data set into power query editor

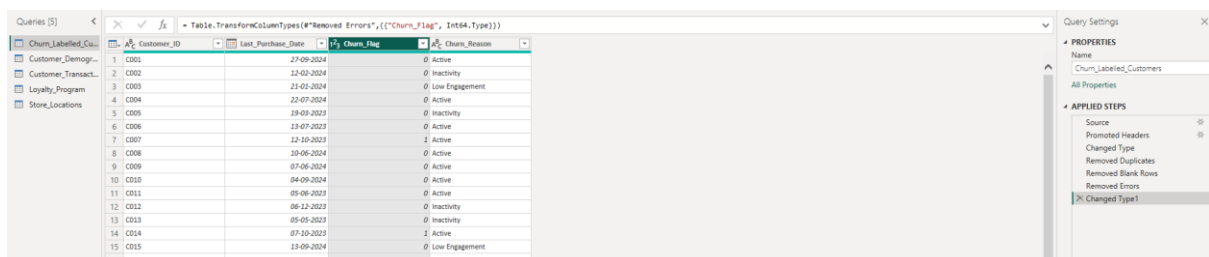
1. Click on get data from home ribbon
2. Select text or CSV
3. Locate your data in your computer
4. Select any one file and open it
5. Press transform data button
6. Now right click on the queries section of the power query editor
Select new query and from the drop down select text or csv
7. Now similarly select other files and load them into the power query editor



Column1	Column2	Column3	Column4
Customer_ID	Last_Purchase_Date	Churn_Flag	Churn_Reason
C001	2024-09-27	0	Active
C002	2024-02-12	0	Inactivity
C003	2024-01-21	0	Low Engagement
C004	2024-07-22	0	Active
C005	2023-03-19	0	Inactivity
C006	2023-07-13	0	Active
C007	2023-10-12	1	Active
C008	2024-06-10	0	Active

- Handle duplicates, missing values, and ensure correct data types

1. After loading the data into the power query editor the first step is to use first row as a headers
2. Then change the data types to the appropriate data types for example the date column should be converted from text data type to date data type. For doing this just select the data type shown in the top left corner of the column. And change it to appropriate one, repeat this type throughout the data sets and make sure all values are set to appropriate data type
3. Now to handle duplicate blank rows and errors just click on remove rows tab on home ribbon and select remove duplicate then repeat the same process and select remove blank rows and then remove errors



Customer_ID	Last_Purchase_Date	Churn_Flag	Churn_Reason
C001	27-09-2024	0	Active
C002	12-02-2024	0	Inactivity
C003	21-01-2024	0	Low Engagement
C004	22-07-2024	0	Active
C005	19-09-2023	0	Inactivity
C006	12-07-2023	0	Active
C007	12-10-2023	1	Active
C008	10-06-2024	0	Active
C009	07-06-2024	0	Active
C010	04-09-2024	0	Active
C011	05-06-2023	0	Active
C012	06-12-2023	0	Inactivity
C013	05-05-2023	0	Inactivity
C014	07-10-2023	1	Active
C015	13-09-2024	0	Low Engagement
C016	19-10-2024	0	Active

- Create calculated columns (Membership_Duration = Today – Membership_Since)

1. After loading and transforming data and handling duplicate missing values and errors click on close and apply on the power query editor home tab

2. Go to the table view and select customer demographic table

3. From home tab add a new column

4. And write this DAX function

Membership_Duration =
DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), MONTH)

Structure

Formatting

Properties

Column

Groups

Relationships

1

Membership_Duration = DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), MONTH)

Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Membership_Duration
C003	34	Female	East	Medium	06 March 2023	Store	33
C009	30	Male	North	Medium	27 December 2019	Online	72
C010	30	Female	Central	Medium	10 June 2019	Online	78
C011	43	Male	West	Medium	19 April 2019	Store	80
C013	59	Male	North	Medium	27 October 2023	Online	26
C017	21	Female	North	Medium	01 June 2020	Online	66
C019	49	Female	East	Medium	23 October 2020	Store	62
C022	40	Female	South	Medium	07 August 2024	Online	16
C024	31	Male	North	Medium	12 March 2022	Online	45
C029	35	Female	North	Medium	07 September 2023	Store	27
C030	34	Male	West	Medium	15 December 2019	Store	72
C031	22	Male	South	Medium	28 November 2020	Store	61
C032	56	Female	East	Medium	31 May 2023	Store	31
C033	26	Male	West	Medium	20 July 2021	Online	52

○ Extract Transaction Year, Transaction Month

For extracting transaction year and transaction month

1. From the home tab go to transform data it will open the power query editor

2. Go into the customer transaction table

3. Click on add column, and select the date option and select the year and repeat the same for the month

Trigonometry	Date	Time	Duration	From Date & Time
Rounding				
Information				

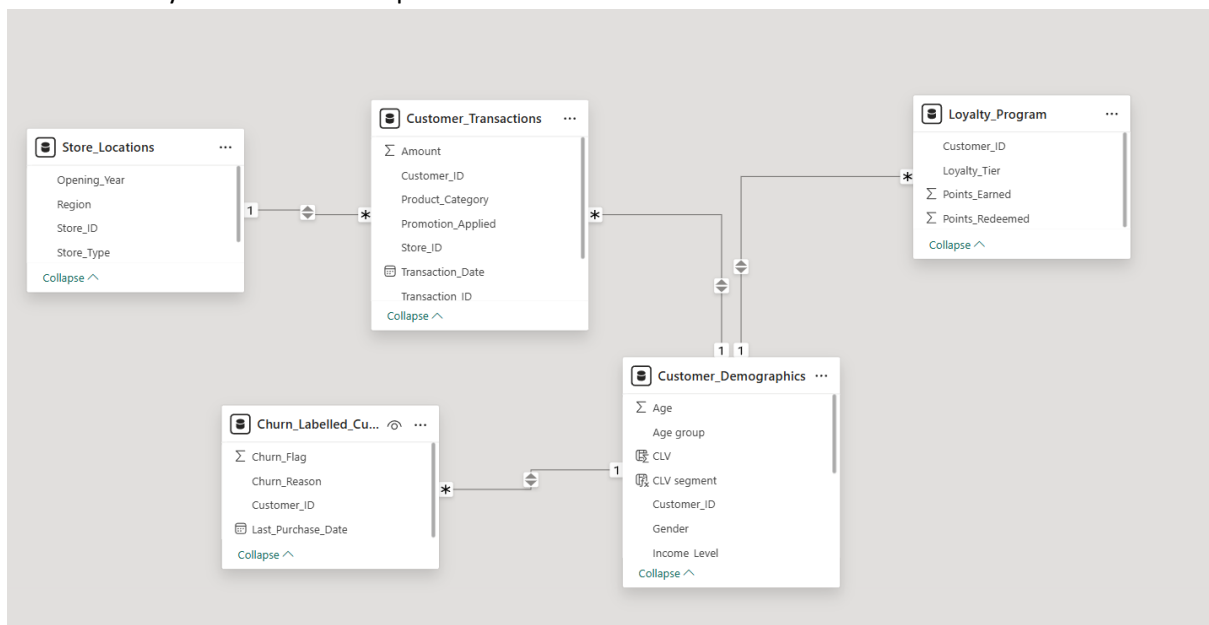
on_Year"}, {"Month", "Transaction_Month"}}}

Transaction_Date	1.2 Amount	A ^B C Promotion_Applied	1 ² 3 Transaction_Year	1 ² 3 Transaction_Month
30-12-2023	129.29	No	2023	12
23-09-2024	788.59	No	2024	9
25-04-2023	419.72	No	2023	4
25-08-2023	333.99	Yes	2023	8
05-10-2023	671.25	No	2023	10
17-03-2024	722.45	No	2024	3
16-03-2023	661.62	No	2023	3
11-02-2023	469.68	Yes	2023	2
17-01-2023	336.09	No	2023	1
10-08-2024	181.01	Var	2024	8

- Creating a basic Data Model view

To create a basic data model view

1. Close and apply the transformed data in power query editor
2. Go to the model view in power bi
3. By analyzing we know that customer demographic is connected with a star schema to customer transaction, loyalty program and churned labelled customer table
4. Drag customer id from customer demographics to customer transactions , select 1 to 2 many relationship from the drop down, and do the similarly to other two tables
5. customer transaction table and Store location are connected to each other through Store ID
6. Drag Store ID from customer transaction to Store ID in store locations table
7. Select many to one relationship



Task 2: Churn & Retention Metrics

- Create Churn Rate KPI = (Churned Customers / Total Customers)

1. For performing the task ahead create a table using enter data from home tab of power bi and name it all measures, this helps to keep all measures organised in one section
2. Right click on the all measures and select enter new measure
3. Now use this dax function to create new measure called total customer

```
Total customer = COUNTA(Customer_Demographics[Customer_ID])
```

4. After that create a new measure called churned customers and use this dax function , this dax function will calculate the Churn customer that are flagged in churn label customers table

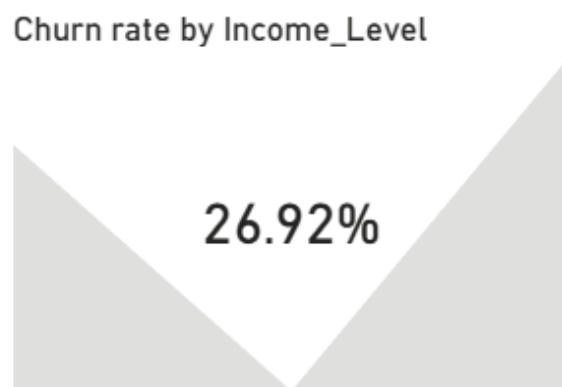
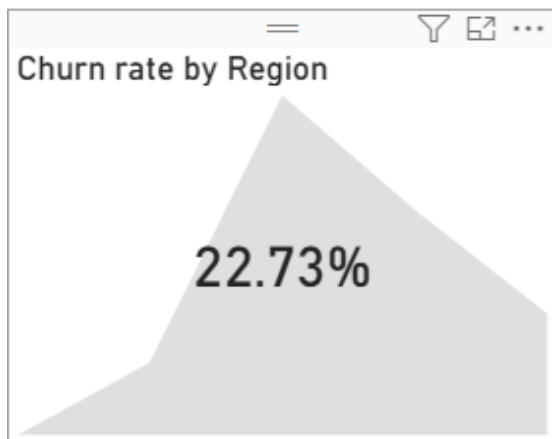
```
Churned Customer =  
CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Chur  
n_Labelled_Customers[Churn_Flag]=1)
```

5. Create a new measure called churn rate and use this dax function

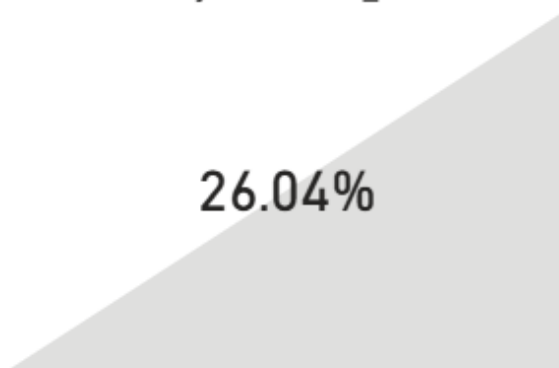
```
Churn rate = DIVIDE([Churned Customer],[Total customer],0)
```

6. Go to the report view in powerbi

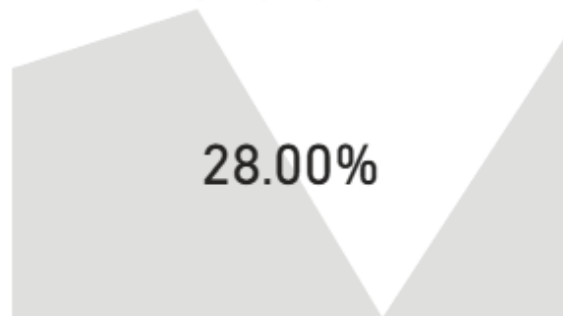
7. Select KPI visual and drag churn rate to value and region in trend axis, repeat this tab for income level ,preferred channel and loyalty tier



Churn rate by Preferred_Channel



Churn rate by Loyalty_Tier



● Funnel Chart:

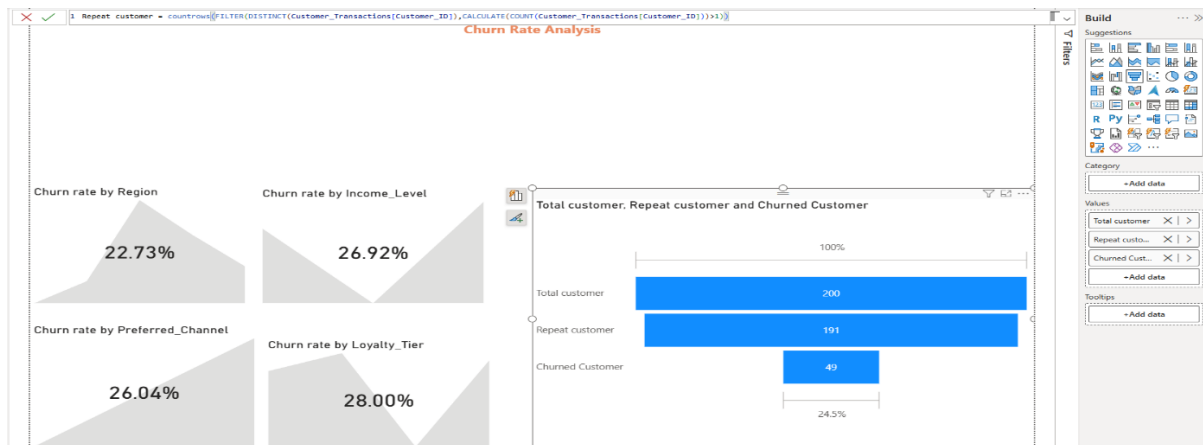
Total Customers → Repeat Customers → Churned

To create a funnel chart that shows total customer, repeat customer and churned customer

1. We have total customer and churned customer measures, we need to calculate repeat customers
2. To calculate this we use measures and use dax function

```
Repeat customer =  
countrows(FILTER(DISTINCT(Customer_Transactions[Customer_ID]),CALCULATE(  
COUNT(Customer_Transactions[Customer_ID]))>1))
```

3. Now that we have all three parameters to create a funnel chart, we can choose funnel chart from build pane and drag all three parameters into values section



Task 3: Repeat Purchase Analysis

- Segment customers:

Low-Tier: 0–3 purchases

Mid-Tier: 4–8 purchases

High-Tier: 9+ purchases

Two segment customers in this specific tiers we have to create a conditional column in power query editor

1. First from the home tab go to transform data and select customer demographics table from the query section
2. In power query editor from add columns tab select conditional column and write the conditions
3. If number of purchase is less than or equal to 3 it should show lower tier else if number of purchase is less than or equal to 8 then mid tier else high tier

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Tier List

	Column Name	Operator	Value ①		Output ①
If	No. of purchase	is less than or equ...	ABC 123 3	Then	ABC 123 Low Tier
Else If	No. of purchase	is less than or equ...	ABC 123 8	Then	ABC 123 Mid Tier

Add Clause

Else ①

ABC 123 High Tier

OK

Cancel

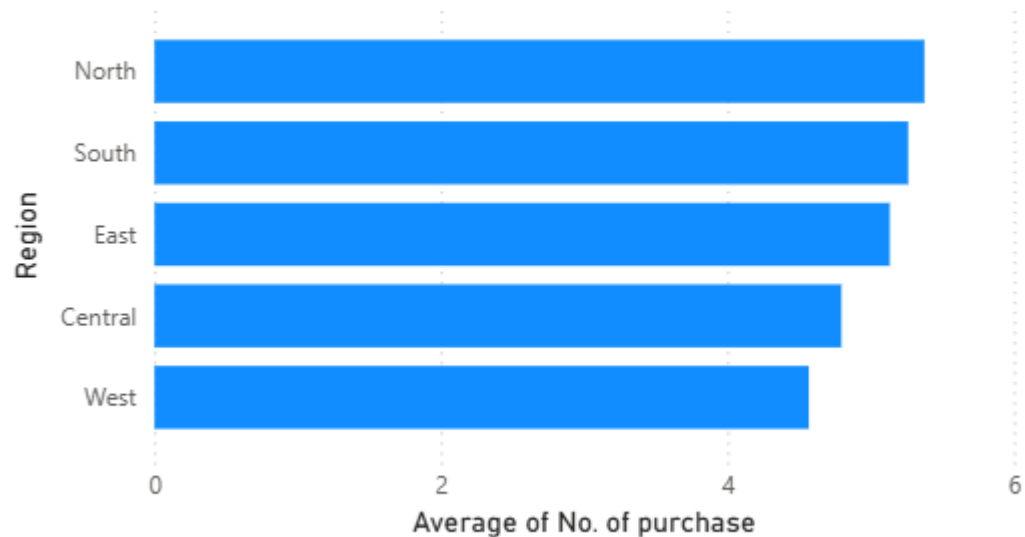
4. now hit close and apply from home tab of power query editor

- Compare avg. purchase frequency by Region, Age Group, Loyalty Tier

To compare average purchase frequency by region age group and loyalty tier, first have to calculate average purchase frequency

1. Now we can visualize average number of purchases by region by creating a clustered bar chart in x axis drag region in y axis drag Number of purchases and from drop down select average

Average of No. of purchase by Region



2. Now to create an average number of purchases by age group we first have to create an age group column in customer demographic table in home select transform data this will open power query editor select customer demographic table from the queries go to add column and select conditional column and use this conditions , now click on close and apply from home tab now go into the report view select clustered column chart on x axis drag age group and on y axis drag Number of purchases and from drop down select average

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Age group

	Column Name	Operator	Value ①		Output ①	
If	Age	is less than or equ...	ABC 123 30	Then	ABC 123 (20-30)	...
Else If	Age	is less than or equ...	ABC 123 40	Then	ABC 123 (31-40)	
Else If	Age	is less than or equ...	ABC 123 50	Then	ABC 123 (41-50)	

Add Clause

Else ①

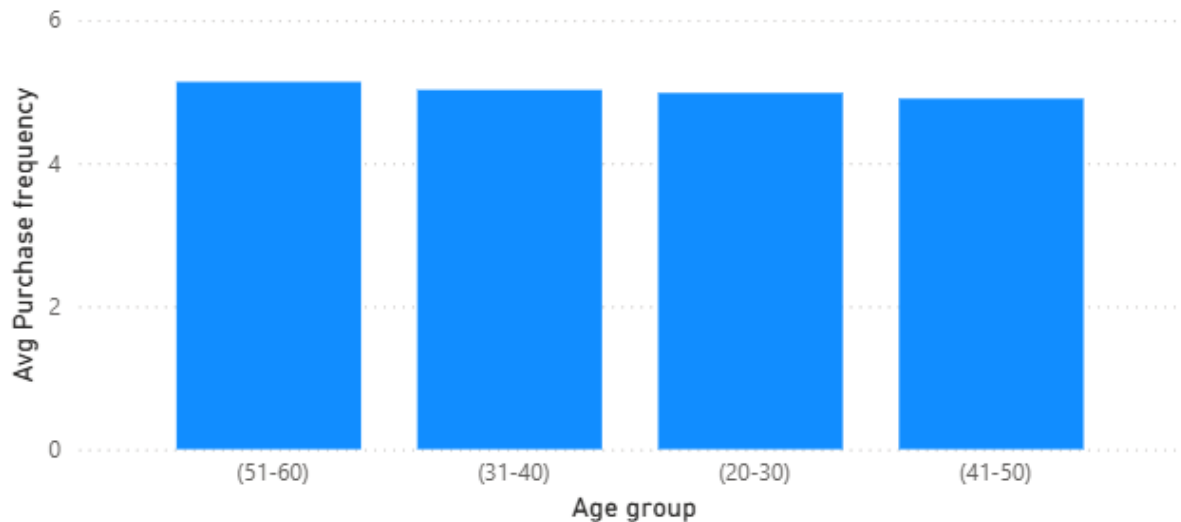
ABC 123

(51-60)

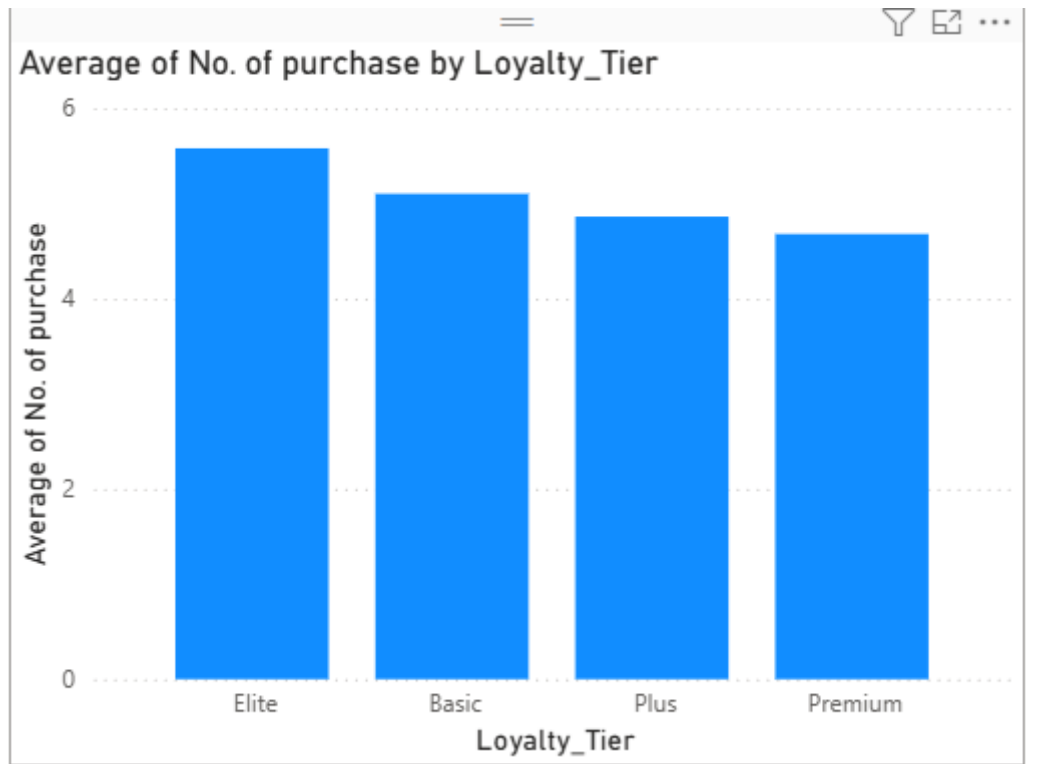
OK

Cancel

Avg Purchase frequency by Age group

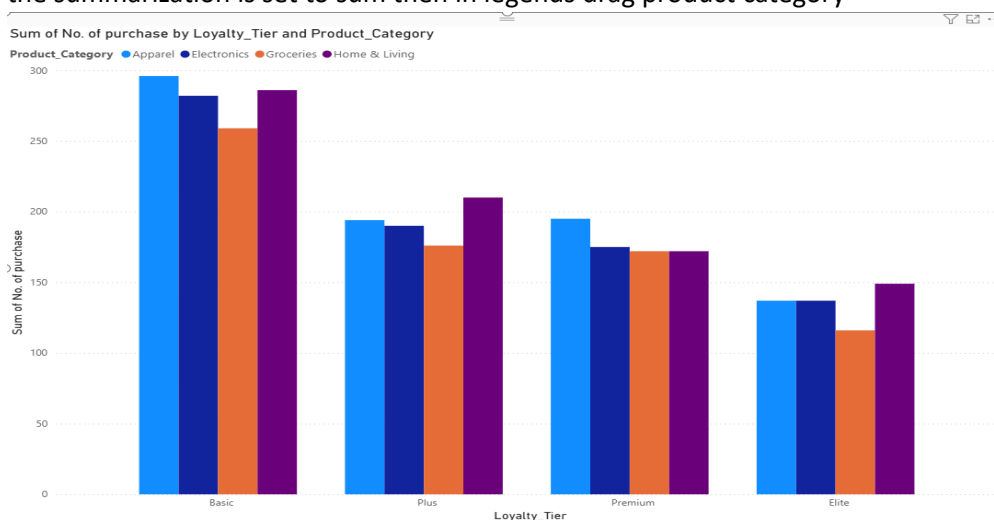


3. To create average purchase frequency by loyalty tier in report view select clustered column chart and drag loyalty tier into x-axis and on y axis drag Number of purchases and from drop down select average



Identify most purchased product categories by loyal customers

To identify most purchase product categories by loyal customers go into report view and create clustered column chart in x-axis drag loyalty tier in y-axis drag number of purchases make sure that the summarization is set to sum then in legends drag product category

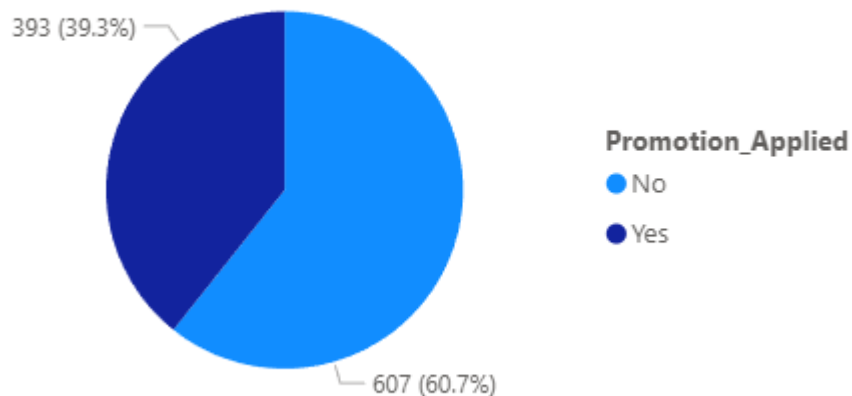


Task 4: Promotion & Loyalty Impact

Calculate % of transactions with promotion applied.

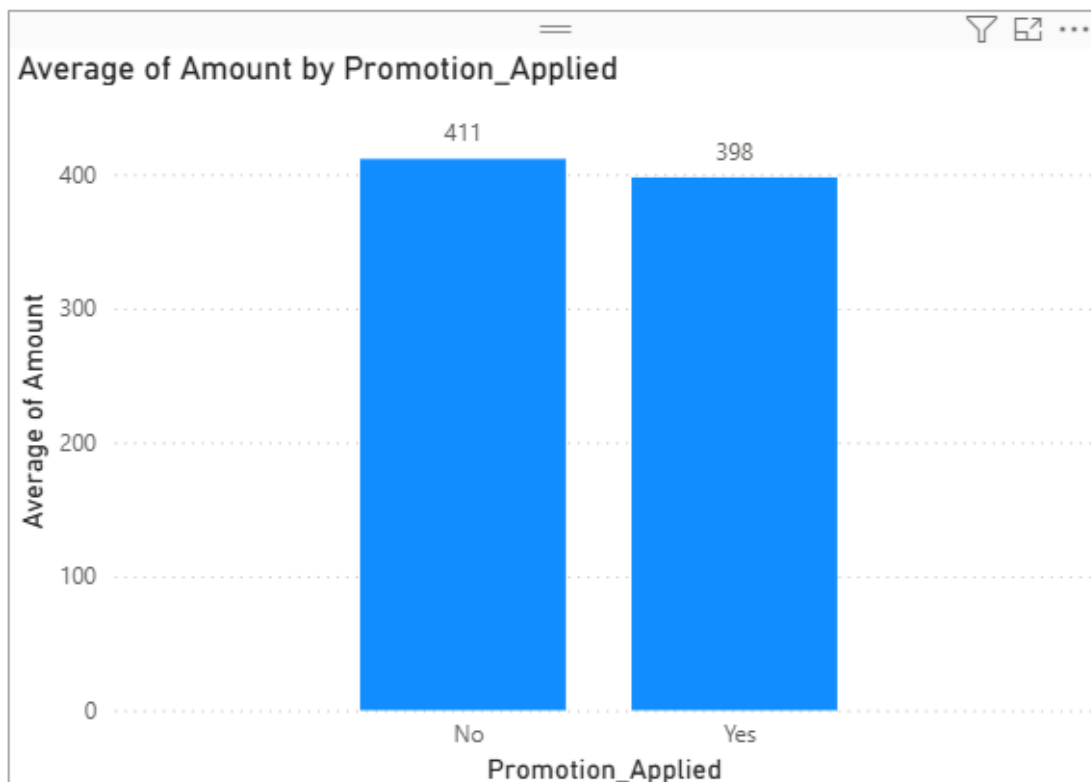
To calculate the percentage we can simply add a pie chart and drag promotion applied column to legends and drag promotion applied to values and change the summarization to count

Count of Promotion_Applied by Promotion_Applied



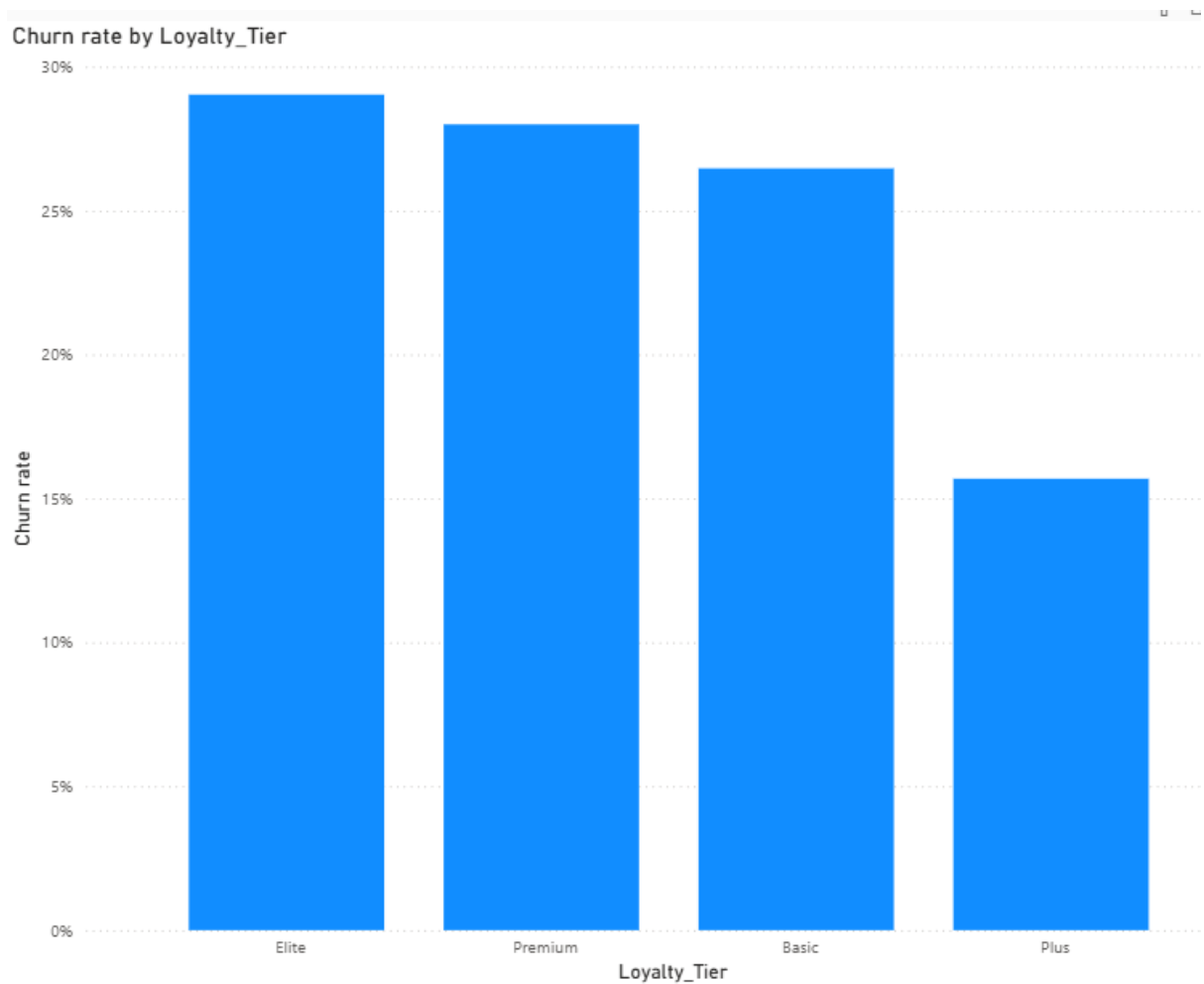
• Compare average purchase amount with vs. without promotions.

To compare average purchase amount with and without promotions we can create a cluster column chart and drag promotion applied on x-axis and amount on y-axis and change the summarization to average



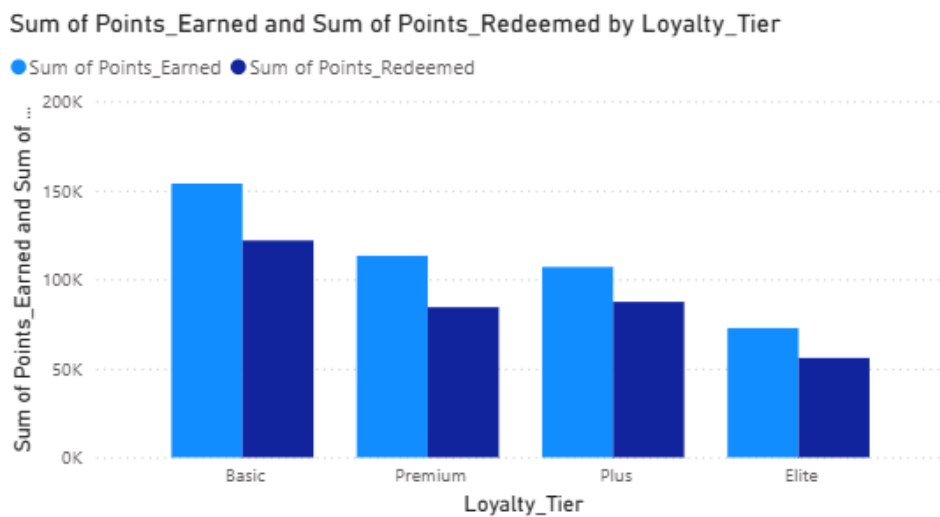
- **Analyze churn rate across loyalty tiers.**

To analyse churn rate by loyalty tier simply create a clustered column chart in report view and drag loyalty tier in x-axis and churn rate in y-axis



- **Visualize Points Earned vs. Redeemed by Tier**

To visualize points earned vs redeemed by tier create a clustered column chart loyalty tier to x-axis and sum of point redeem and sum of point earned to y-axis



- Give recommendations to improve redemption & retention.

To improve redemption of points earned Target should provide more than discount coupons as we see in the results Average Amount spend by promotion applied is almost similar to when the promotion are not applied , one suggestion would be to increase the value of points so customers would spent them more , second recommendation would be to give customers from higher tier more points than basic tier as they are our valuable customer , here in trade of points they can have exclusive product available or they can have perks like first try on for a specific product

To improve retention we have to focus more on customers at high risk of churning that is customer with lower engagement through my analysis they are mostly medium in come level and from age 20 - 30 , running more online campaigns and providing more offers might stop the from churning and increase retention

Task 5: Store & Channel Performance vs Retention

- Merge store data with transactions

To merge store data with transactions go to transform data in queries select store locations table select merge queries from home tab and from drop down select customer transactions and select Store ID from both table , now click on expand button and select aggregate and select some of amount now just change the data type to decimal number and click on close and apply

×

Merge

Select a table and matching columns to create a merged table.

Store_Locations

Store_ID	Store_Type	Region	Opening_Year
S01	Online	East	2018
S02	Neighborhood Market	West	2017
S03	Online	West	2009
S04	Online	Central	2010
S05	Online	Central	2014

Refresh

Customer_Transactions

Transaction_ID	Customer_ID	Store_ID	Product_Category	Transaction_Date	Amount	Promotion_Appl
T0001	C051	S17	Home & Living	30-12-2023	129.29	No
T0002	C072	S06	Home & Living	23-09-2024	788.59	No
T0003	C110	S08	Apparel	25-04-2023	419.72	No
T0004	C044	S19	Groceries	25-08-2023	333.99	Yes
T0005	C003	S10	Home & Living	05-10-2023	674.05	No

< >

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

OK

Cancel

Search Columns to Expand



☐ Expand ☒ Aggregate

☒ (Select All Columns)

☐ # Count of Transaction_ID

☐ # Count of Customer_ID

☐ # Count of Store_ID

☐ # Count of Product_Category

☐ # Count of Transaction_Date

☒ Σ Sum of Amount

☐ # Count of Promotion_Applied

☐ Σ Sum of Transaction_Year

☐ Σ Sum of Transaction_Month

☒ Use original column name as prefix

OK

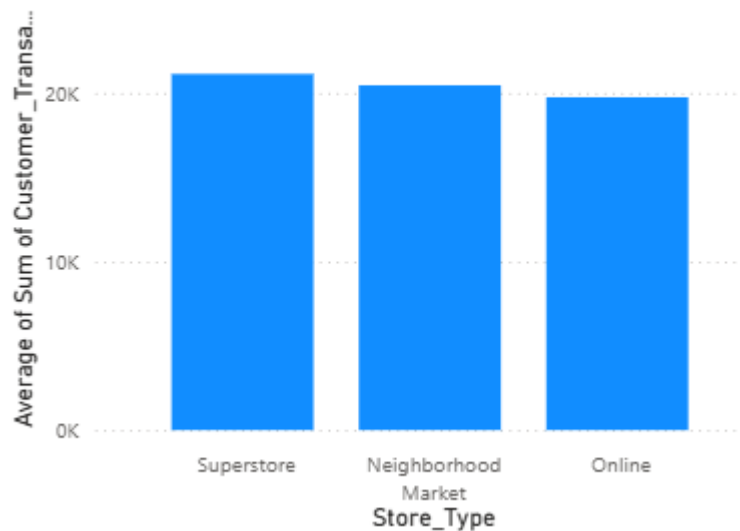
Cancel

	A _C Store_ID	A _C Store_Type	A _C Region	1 ₂ Opening_Year	1.2 Sum of Customer_Transactions.Amount
1	S01	Online	East	2018	17157.89
2	S02	Neighborhood Market	West	2017	24190.99
3	S03	Online	West	2009	24067.34
4	S04	Online	Central	2010	18055.99
5	S05	Online	Central	2014	16525.36
6	S06	Online	South	2014	20734.79
7	S07	Superstore	South	2019	20247.9
8	S08	Neighborhood Market	North	2011	20823.09
9	S09	Superstore	Central	2012	21917.11
10	S10	Superstore	East	2012	20387.22
11	S11	Neighborhood Market	Central	2013	19630.41
12	S12	Neighborhood Market	East	2021	19097.36
13	S13	Superstore	West	2016	21958.1
14	S14	Neighborhood Market	Central	2009	23651.11
15	S15	Online	West	2012	13789.37
16	S16	Neighborhood Market	South	2019	20697.09
17	S17	Neighborhood Market	South	2010	17386.08
18	S18	Online	North	2011	22298.57
19	S19	Online	Central	2017	25305.96
20	S20	Neighborhood Market	West	2013	18148.58

- **Visualize Avg. transaction amount by Store Type**

Two visualize average transaction amount by stone type go to report view select a clustered column chart in x axis drag store type and in y axis drag sum of customer transaction column just added and change the summarization to average

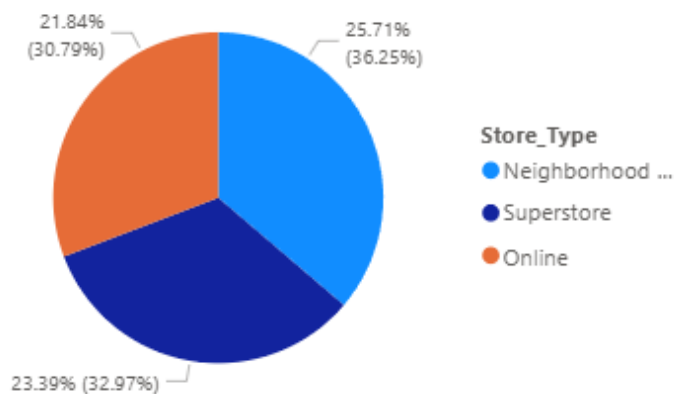
Average of Sum of Customer_Transactions.Amount by Store_Type



- **Churn rate by store type**

To calculate churn rate by store type go to the report view and create a pie chart in legends drag store type and in values drag churn rate

Churn rate by Store_Type

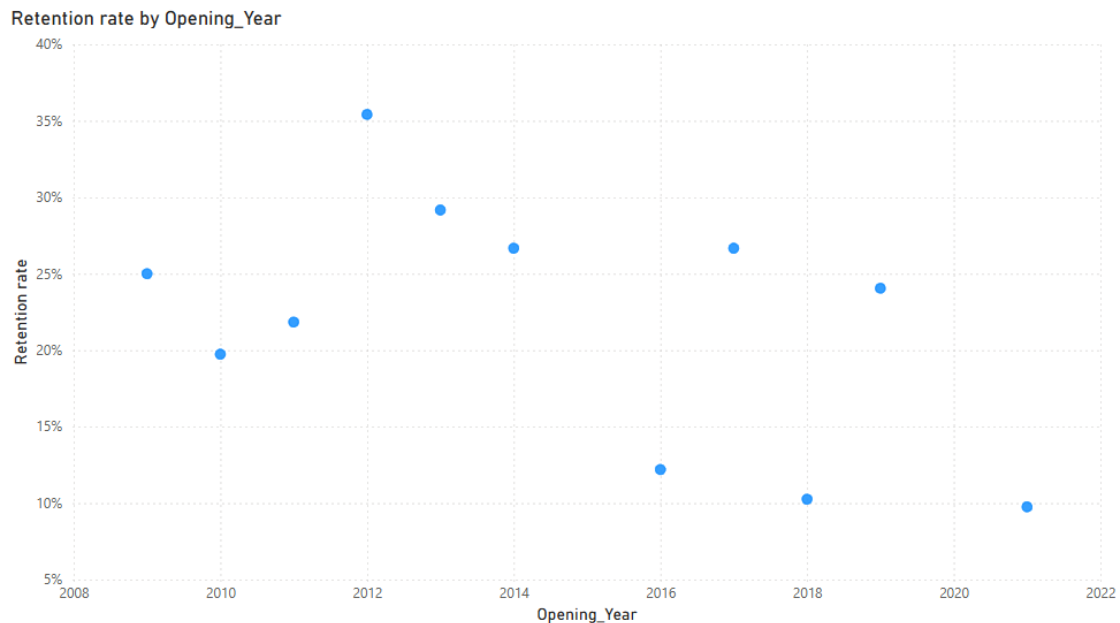


○ Correlation between store opening year & retention

To visualize correlation between store opening and retention first we have to calculate retention rate create a new measure in all measures and use this dax function it will divide repeat customer with total customer this will give us the retention rate

$$\text{Retention rate} = \text{DIVIDE}([\text{Repeat customer}], [\text{Total customer}], 0)$$

then to find correlation the best chart is scatter plot go to the report view and select scatter plot from build pane in x-axis drag opening year and in y axis drive retention rate



Task 6: Customer Lifetime Value (CLV) Analysis

To calculate customer lifetime value we first have to calculate total amount spent by each customer

1. In home tab go to transform data from query select customer demographic now select merge queries and select customer transactions select customer ID in both table
2. Then using drop down select aggregate and aggregate using sum of amount
3. Now the data can have nulls replace the null values with zero using replace values in home tab
4. Change the type to decimal values hence the total amount spend table is created
5. Now hit close and apply and in table view go to customers demographic table and add column using calculated column create clv

$$\text{CLV} = \text{DIVIDE}(\text{Customer_Demographics}[\text{Total amount spend}], \text{Customer_Demographics}[\text{Membership_Duration}], 0)$$

Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Membership_Duration	No. of purchase	Tier List	Age group	Total amount spend	CLV
C001	58	Female	North	Low	18 August 2023	Online	2	2	7 Mid Tier	(51-60)	3819.76	1909.9
C002	48	Male	North	High	17 September 2019	Store	6	6	5 Mid Tier	(41-50)	1327.71	221.3
C005	40	Male	South	High	12 September 2022	Online	3	3	5 Mid Tier	(31-40)	1764.29	588.1
C006	58	Male	South	High	30 October 2018	Online	7	7	6 Mid Tier	(51-60)	852.48	121.8
C007	38	Female	East	High	01 October 2019	Store	6	6	7 Mid Tier	(31-40)	3365.44	560.9
C010	30	Female	Central	Medium	10 June 2019	Online	6	6	4 Mid Tier	(20-30)	2111.99	352.0
C012	55	Female	South	High	28 January 2018	Store	7	7	5 Mid Tier	(51-60)	2054.71	293.5
C015	22	Male	Central	High	10 June 2019	Online	6	6	7 Mid Tier	(20-30)	1602.31	267.1
C016	41	Female	West	High	28 August 2021	Online	4	4	6 Mid Tier	(41-50)	2553.68	638.4
C017	21	Female	North	Medium	01 June 2020	Online	5	5	6 Mid Tier	(20-30)	2361.56	472.3

● Segment customers into Low, High CLV

We need to create a measure that calculates average clv

average clv = AVERAGE(Customer_Demographics[CLV])

In table view select customers demographic table go to table tools add column and write if tax function that compare if customer clv is higher than average it displays high clv or else low clv

CLV segment = IF(Customer_Demographics[CLV] >= CALCULATE([average clv],ALL(Customer_Demographics)), "High CLV" , "Low CLV")

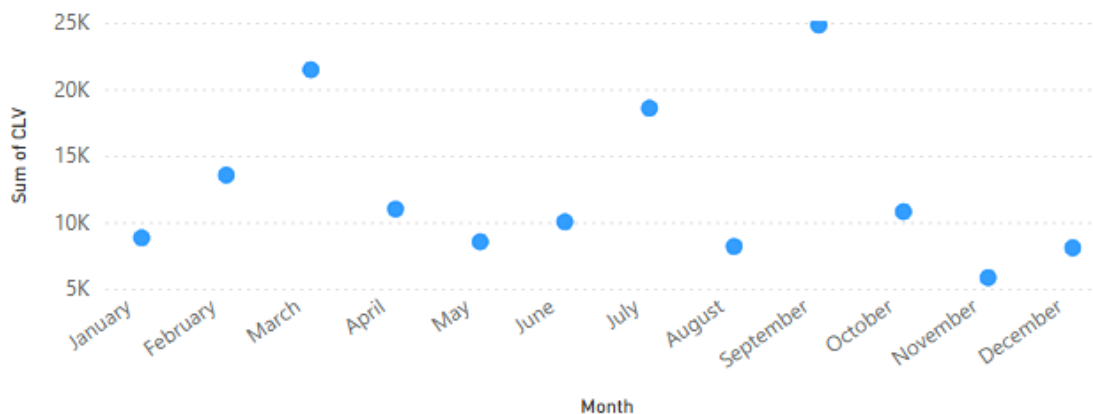
1	CLV segment = IF(Customer_Demographics[CLV] >= CALCULATE([average clv],ALL(Customer_Demographics)), "High CLV" , "Low CLV")
---	---

Customer_ID	Age	Gender	Region	Income_Level	Membership_Since	Preferred_Channel	Membership_Duration	No. of purchase	Tier List	Age group	Total amount spend	CLV	CLV segment
C001	58	Female	North	Low	18 August 2023	Online	2	2	7 Mid Tier	(51-60)	3819.76	1909.9	High CLV
C002	48	Male	North	High	17 September 2019	Store	6	6	5 Mid Tier	(41-50)	1327.71	221.3	Low CLV
C005	40	Male	South	High	12 September 2022	Online	3	3	5 Mid Tier	(31-40)	1764.29	588.1	Low CLV
C006	58	Male	South	High	30 October 2018	Online	7	7	6 Mid Tier	(51-60)	852.48	121.8	Low CLV

○ CLV vs Days Since Last Purchase

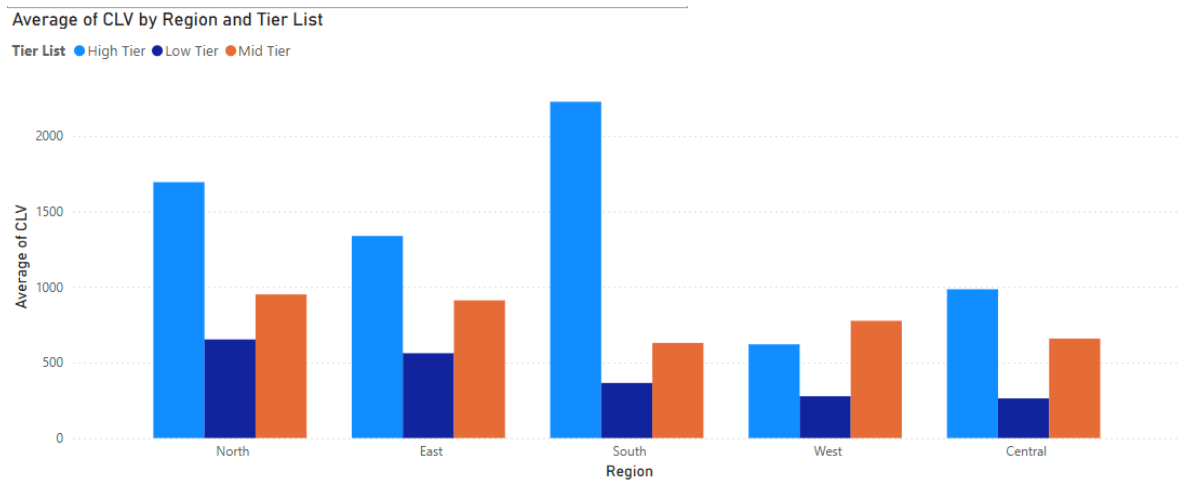
To visualise clv since last day of purchase in report view select a scatter plot on x-axis drag last purchase and keep only month section and in y axis drag sum of clv

Sum of CLV by Month



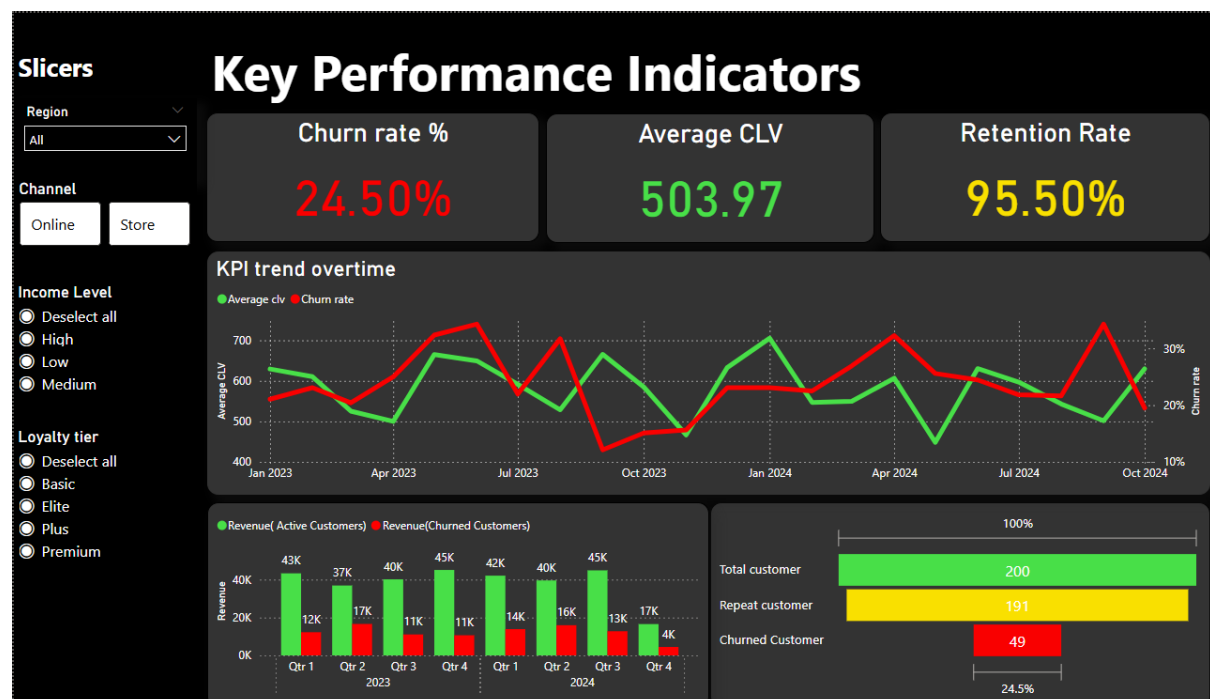
○ CLV by Loyalty Tier & Region

To visualise this, in report view select clustered column chart drag region to x axis and clv to y axis change the summarization to average.



Task 7: Final Dashboard & Executive Summary

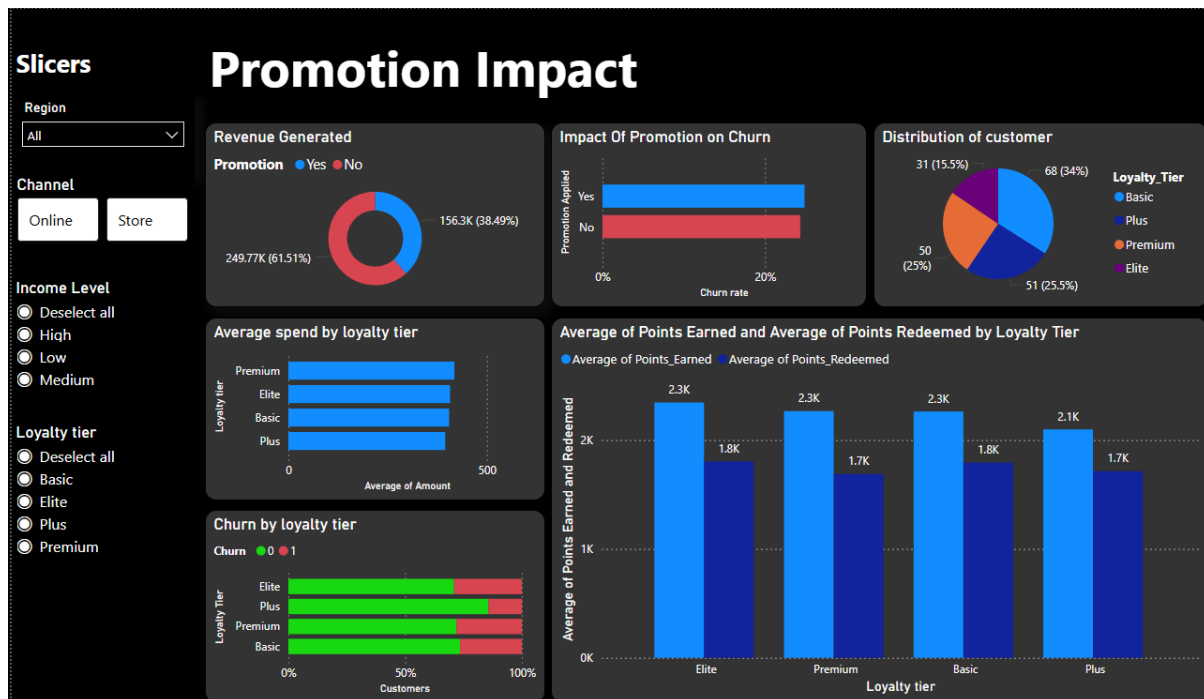
○ Page 1: KPIs (Churn, CLV, Repeat Rate)



Key findings and recommendations

1. As average clv rises churn rate decreases and when average clv increases churn rate decreases
2. Growth is stagnant due to high churn rate
3. Customers with repeat orders are also churning

○ Page 2: Loyalty & Promotion Impact



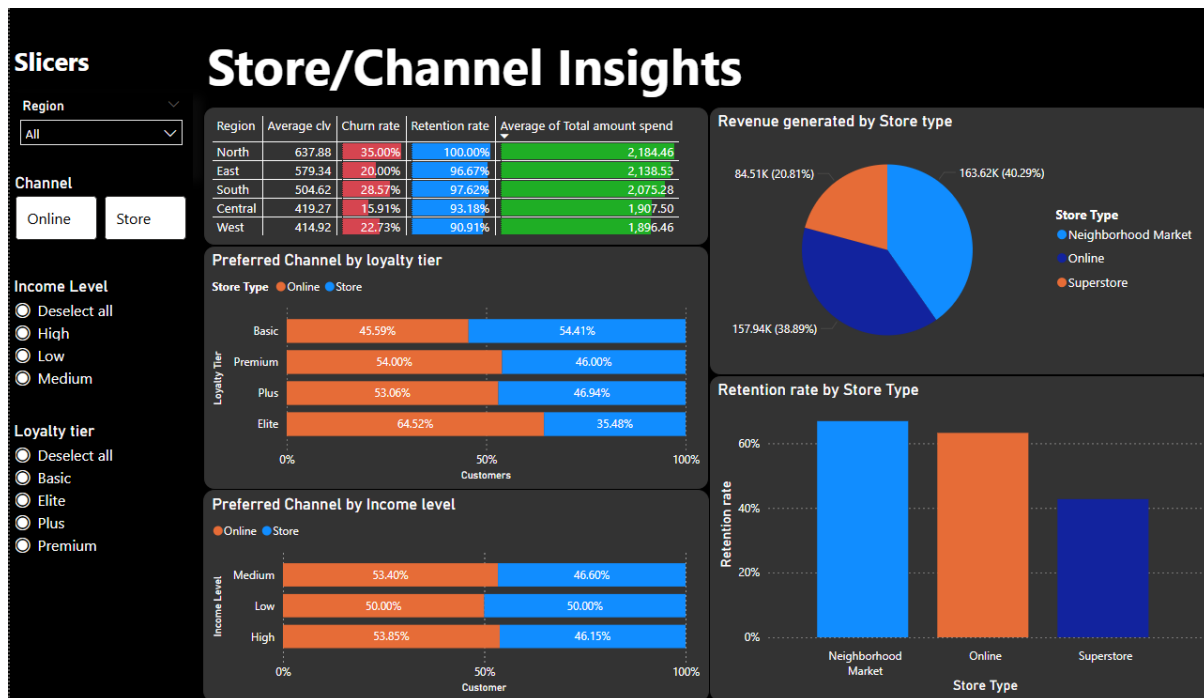
Key findings

1. Customer that apply promotion during order placement tend to churn more
2. Elite tier customers on average earn similar points as basic and premium tier

Recommendation

1. Allocate more points to customer elite tier
2. Reduce amount of point basic tier customer
3. Make sure elite tier customer get more perks as compared to other tiers as they are more valuable customers
4. Increase perks for elite tier to reduce churn

○ Page 3: Store/Channel Insights



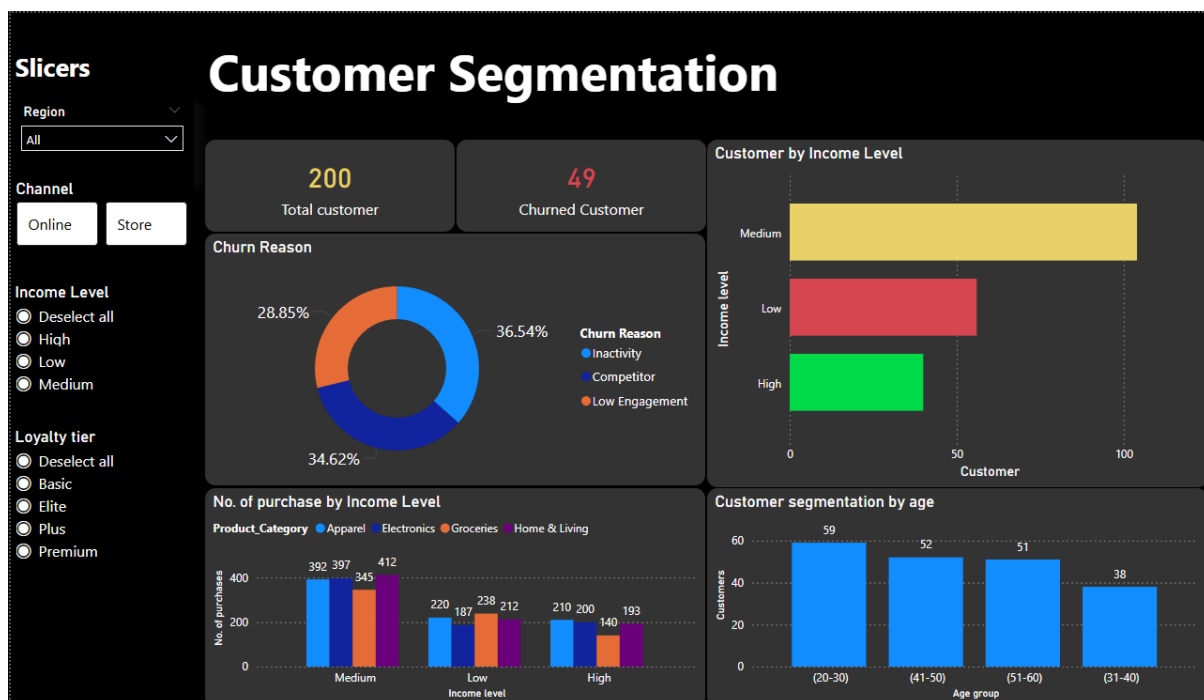
Key findings

1. SuperStores generate less revenue and also has lowest retention rate
2. Neighborhood market have highest retention and revenue
3. The preferred channel for high value customers is online channel

Recommendation

1. Focus more on online and neighborhood markets as they have less upfront cost and also generate more revenue and has more retention
2. Make online Channels way more accessible and user friendly as more high value customers prefer this channel

○ Page 4: Segmentation



Key findings and Recommendation

As the churn reason for customer is low engagement and competitor we can revive these customers running online campaigns targeting customer with mid and high level income , focus more on specific product category and target age group of 20-30 years

- Slicers: Region, Channel, Income, Loyalty Tier

These slicers were added at left throughout the dashboard for easy analysis

- Provide Top 3 Recommendations for Target:

○ Which customers to prioritize for retention?

Customer with elite and plus tier , from north region

○ Which channels are underperforming?

Superstore are under performing with less revenue and retention

○ How to strengthen loyalty engagement?

Increase benefits and perks for elite tier customers also allocate more points to this customers

Task 8: Video explanation: Expressing the finding and actionable insights

Video link

https://drive.google.com/file/d/1Lm1tPM6v7y-yDYNrnZMv8PNMbiDMZ9_d/view?usp=sharing